DOCUMENT 05 OF 06 • DATA ENGINEERING & API SPECIFICATION

PostgreSQL Schema, REST/WebSocket Contracts & Token Ledger

Database: PostgreSQL 16 + pgvector | **Transport:** REST (JSON) + WebSockets (Live Agent Telemetry)

1. Core PostgreSQL Database Schema (DDL Blueprint)

The database schema enforces strict financial accounting for the white-labeled OpenRouter proxy while storing versioned Master CVs, search missions, morphed resumes, and interview defense dossiers.

```
-- 1. USERS & MASTER CAREER KNOWLEDGE GRAPH
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email TEXT UNIQUE NOT NULL,
    full_name TEXT NOT NULL,
    master_cv_raw_text TEXT,
    master_knowledge_graph JSONB NOT NULL DEFAULT '{}'::jsonb, -- Verified companies,
dates, base skills
    cv_embedding vector(1536),                                  -- For instant semantic
job matching
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- 2. SUBSCRIPTION & WHITE-LABEL AI WALLET
CREATE TABLE user_wallets (
    user_id UUID PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE,
    subscription_status TEXT NOT NULL DEFAULT 'INACTIVE',      -- 'ACTIVE',
'GRACE_PERIOD', 'EXPIRED'
    subscription_plan_id TEXT,                                  -- 'apex_monthly_399'
    subscription_expires_at TIMESTAMPTZ,
    credit_balance_usd NUMERIC(12, 6) NOT NULL DEFAULT 0.500000, -- Includes $0.50
welcome bonus
    preferred_routing_mode TEXT NOT NULL DEFAULT 'MANAGED',    -- 'MANAGED' or 'BYOK'
    preferred_model_slug TEXT NOT NULL DEFAULT 'apex-pro',
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- 3. IMMUTABLE LLM TOKEN & +$1/1M MARKUP LEDGER
CREATE TABLE llm_usage_ledger (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    mission_id UUID,
    execution_mode TEXT NOT NULL,                               -- 'MANAGED_PROXY' or
'BYOK'
    upstream_model_slug TEXT NOT NULL,                          -- e.g.,
'anthropic/claude-3.7-sonnet'
    prompt_tokens INT NOT NULL,
    completion_tokens INT NOT NULL,
    upstream_openrouter_cost_usd NUMERIC(12, 6) NOT NULL DEFAULT 0,
    platform_markup_usd NUMERIC(12, 6) NOT NULL DEFAULT 0,     -- (Total Tokens / 1M) *
$1.00
    total_deducted_usd NUMERIC(12, 6) NOT NULL DEFAULT 0,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- 4. AUTONOMOUS SEARCH MISSIONS
CREATE TABLE search_missions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    mission_mode TEXT NOT NULL,                                 -- 'SALARIED_JOB' or
'FREELANCE_GIG'
    target_countries TEXT[] NOT NULL,                           -- e.g., '{US, GB, AE,
REMOTE}'
    commitment_types TEXT[] NOT NULL,                           -- e.g., '{FULL_TIME,
```

2. REST & WebSocket API Contract Specification

Method & Endpoint	Headers / Payload	Response & Behavior
`GET /api/v1/llm/catalog`	`Authorization: Bearer <jwt>`	Returns white-labeled model catalog with OpenRouter branding stripped and `+$1.00/1M` token markup pre-applied to display rates.
`POST /api/v1/profile/parse-cv`	`Multipart/Form-Data (file)` + Optional `X-BYOK-Key`	Extracts text from PDF/DOCX, constructs `master_knowledge_graph` JSON, and computes initial vector embedding.
`POST /api/v1/missions/launch`	JSON: `mode`, `countries[]`, `commitment[]`, `keywords[]`, `aggressiveness`	Enqueues Celery discovery + tailoring pipeline. Returns `mission_id` and WebSocket channel URL for live telemetry.
`POST /api/v1/tailor/single-jd`	JSON: `raw_jd_text`, `mode`, `aggressiveness` + Optional `X-BYOK-Key`	Immediate single-target execution: runs Gap Audit, Chameleon CV Rewrite, Typst PDF compile, and Sensei Prep generation in one call.
`WS /ws/v1/missions/{id}`	WebSocket Upgrade with JWT token	Pushes real-time JSON events: `JOB_FOUND`, `GAP_DETECTED`, `SKILL_GRAFTED`, `PDF_READY`, `WALLET_UPDATED`.
`POST /api/v1/billing/webhook`	RevenueCat / Razorpay HMAC Signature	Handles `INITIAL_PURCHASE`, `RENEWAL` (unlocks monthly subscription), and `NON_RENEWING_PURCHASE` (adds $5/$10 to `credit_balance_usd`).

3. JSON Contract: `injected_skills_delta` & `sensei_prep_dossier`

This exact JSON payload powers both the green diff highlights in the Flutter CV Viewer and the flashcards inside the Skill-Mimic Dojo tab:

```
{
  "artifact_id": "9f8c2b10-44a1-4e9b-8a1d-32891c7b90e1",
  "base_ats_score": 54.0,
  "final_ats_score": 97.5,
  "injected_skills_delta": [
    {
      "skill_name": "Apache Kafka & Event-Driven Microservices",
      "jd_importance": "MANDATORY",
      "grafted_into_role": "Backend Developer @ TechNova Solutions (2023 - Present)",
      "original_bullet": "Built REST APIs in Node.js for order processing and database
updates.",
      "morphed_bullet": "Architected high-throughput event-driven order pipelines using
Apache Kafka and Node.js with idempotent consumers and dead-letter queues, reducing
peak checkout latency by 42%.",
      "sensei_defense": {
        "sixty_second_mental_model": "Kafka is a distributed commit log where producers
write events to partitioned topics and consumer groups read them asynchronously without
tight coupling.",
        "star_story_script": "At TechNova, synchronous REST calls were timing out
during flash sales. I introduced a 3-broker Kafka cluster partitioned by `merchant_id`
so order events were ingested in under 15ms and processed asynchronously by inventory
workers...",
        "trap_questions": [
          {
            "question": "How did you prevent duplicate order processing when a Kafka
consumer crashed mid-batch?",
            "golden_answer": "We disabled auto-commit and committed offsets manually
only after writing the order state and event `idempotency_key` inside a single
PostgreSQL transaction."
          }
        ]
      }
    }
  ]
}
```